Denomination, Not Depth: Cross-Sectional Cascade Exposure in DeFi Lending

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Abstract

We study how price triggered liquidations responses are distributed across decentralized finance (DeFi) lending markets. Using a heterogeneous Panel SVAR in the framework of Pedroni (2013), estimated separately for each of 38 protocol–chain pairs over July 2024–December 2025, we find that a -1% same-day shock to the collateral basket raises USD liquidation volume by roughly 22% on impact and 28% one day later, decaying within five days. Two cross-sectional second-stage regressions then ask which features of a market predict its response. Market depth, measured by total value locked, has no detectable effect ($\hat{\beta} = -0.008$, $p = 0.27$). Debt denomination does: regressing the per-CSU common-component IRF on the stablecoin share of the debt book yields $\hat{\beta} = -0.31$ ($t = -7.0$, $R^2 = 0.54$), so a fully stablecoin-debt market translates a -1% collateral shock into roughly 36% more USD liquidation than a fully volatile-debt market. Cascade exposure in DeFi lending therefore concentrates along the denomination dimension rather than along market size. The analysis is supported by an open-source multi-chain data pipeline covering 23 protocol families across 17 blockchains.

Contents

1 Introduction

Decentralized Finance (DeFi) enables users to lend, borrow, and trade digital assets through smart contracts—self-executing code on the blockchain—without intermediaries. To protect lenders in this pseudonymous setting, borrowers must post collateral worth more than their loan. If collateral value falls below a protocol-defined threshold, the position is liquidated: collateral is seized and sold to repay lenders.

The link from collateral-price declines to liquidation volume has been established empirically. What is not as well understood is how collateral price declines impact liquidations across different protocols. DeFi lending is not one market but several dozen, each with its own borrow book, collateral basket, and debt denomination and how exposure is distributed uniformly across these markets, or concentrates along particular institutional dimensions, is the primary focus of this paper.

We estimate a heterogeneous Panel SVAR separately for each of 38 protocol–chain pairs over July 2024 to December 2025. The bivariate system [collateral basket return , log-liquidation] uses a Cholesky orthogonalization in which the liquidation shock has zero same-day effect on the basket return. The panel-wide common-component impulse response shows that a -1% same-day collateral-return shock raises USD liquidation by roughly 22% on impact and 28% one day later, decaying within five days. Two cross-sectional second-stage regressions then ask which features of a market predict its response. Market depth, measured by total value locked, has no detectable effect on the price-triggering response. Debt denomination does: regressing the per-CSU common-component IRF at $h = 1$ on the count-weighted stablecoin share of the debt book yields $\hat{\beta} = -0.31$ ($t = -7.0$, $R^2 = 0.54$), so a fully stablecoin-debt market translates a -1% collateral shock into roughly 36% more USD liquidation than a fully volatile-debt market. Shock exposure in DeFi lending therefore concentrates along the denomination dimension rather than along market size.

A portion of the contribution is methodological. On-chain DeFi data are publicly available in principle but difficult to use in practice, requiring protocol-specific decoding, multi-chain infrastructure, and event-log expertise. We assemble an open-source data pipeline organized around a protocol-adapter pattern that covers 23 protocol families across 17 blockchains and produces the balanced analysis-ready panels used below.

2 Literature Review

Our theoretical and empirical understanding of DeFi lending markets has greatly increased in the past five years (2021). Prior to that period, the best understanding of the fire sale externality present in DeFi came from the traditional finance mortgage market. The fire sale externality, or cascading liquidation dynamic in DeFi, is almost the same feedback Geanakoplos's leverage cycle describes: forced sales of seized collateral push prices down, lower prices push more positions across their thresholds, and the loop amplifies small shocks into larger ones (Geanakoplos, 2010). Building off that and other foundations in traditional finance, Chiu et al. (2023) formalize the DeFi-specific version of this price-liquidity feedback. Unlike other areas of financial economics, the transparent and deterministic nature of smart contracts makes the mechanisms underlying DeFi lending dynamics unambiguous.

Empirically, DeFi is greatly understudied relative to traditional finance analogs, but it has seen some major contributions recently. Perez et al. (2021) captures the vulnerability of DeFi lending, finding that a 3% price decline makes more than $10 million of Compound positions liquidatable and that over 70% of liquidations occur within the same block as the triggering oracle update. Lehar & Parlour (2022) also document liquidation-induced price impacts on Aave and Compound through large event waves impacting DEX liquidity. These and others (OECD, 2023) provide evidence that cascading liquidations can exist in DeFi, that they happen at block-level speed, and that their impact can range from essentially zero to millions of dollars.

Considering different mechanism designs, Tian & Zhu (2025) find that auction-based liquidation (MakerDAO) produces meaningfully smaller price impacts than fixed-spread liquidation (Aave), but overall, cross-protocol variation is largely unexplored.

The impact of leverage on causing liquidations was addressed by Heimbach & Huang (2024), who find that higher leverage raises the share of positions near liquidation thresholds but does not significantly predict actual liquidation outcomes. Essentially, that leverage makes it easier for collateral price shocks to cause a liquidation, and that liquidations are event-driven by collateral-price shocks.

Most recently, Chiu & Danisman (2026) from the Bank of Canada decompose the significance of the different possible changes to the health factor, finding that changes to the collateral price account for 84-97% of health factor deteriorations across Aave V3 on Ethereum. Additionally, using a FE model, they find no statistically significant evidence that liquidation activity leads to a persistent effect on market prices. Together, these papers establish the microfoundation and existence of cascading liquidations at the block or intra-block level that are triggered by collateral asset price changes, vary across different protocol designs, and do not have significant effects when aggregated to the daily level.

2.1 Contribution to the Literature

The paper is the first, to our knowledge, to use a PVAR (Panel Vector Autoregression) approach to deconstruct DeFi lending and to explore heterogeneity across protocols and chains using a wider set of lending markets. Second, a portion of the contribution is methodological. On-chain DeFi data is in principle open but in practice hard to work with, so we assemble an open-source pipeline organized around a protocol-adapter pattern that covers 23 protocol families across 17 blockchains and produces balanced analysis-ready panels from raw blockchain state.

3 Conceptual Framework

DeFi lending protocols allow users to deposit digital assets as collateral and borrow other assets against them. Because all participants are pseudonymous, overcollateralization—requiring collateral worth more than the loan—protects lenders, and smart contracts enforce the terms of the loan without recourse beyond the collateral itself. If the position becomes undercollateralized, the contract authorizes its liquidation and the collateral is sold to repay the lender. A numerical walkthrough of the associated leverage, liquidation, and price amplification mechanism is given in Appendix A; the remainder of this section sets up the institutional features we use to interpret the empirical analysis. We begin with the health-factor ratio that determines each individual liquidation.

3.1 The health factor

Whether a loan becomes liquidated is determined based on a health factor. For a position with collateral of market value V_C in USD, debt of market value V_D, and a protocol-defined liquidation threshold θ,

$$HF = \frac{V_C \cdot \theta}{V_D}.$$

The position is safe when $HF \geq 1$ and liquidatable when $HF < 1$. Writing $V_C = P^C Q^C$ and $V_D = P^D Q^D$, the log change of the health factor, ignoring intra-day rate and quantity adjustments, reduces to

$$\Delta \log HF \approx r^C - r^D,$$

where r^C is the return on the collateral and r^D the return on the debt.

In §6.4 we test whether the level of stablecoin denominated debt in a lending market explains that market's response to a common shock to the price of the collateral asset. This is motivated by the fact that when the debt is a stablecoin, $r^D \approx 0$, and any full collateral shock should pass through directly to the health factor. Alternatively, when the debt is itself a volatile asset correlated with the collateral, the two returns partially offset and the health factor is comparatively insulated from broad market moves.

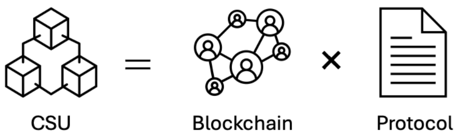
3.2 How a lending protocol works

At the more macro level is a lending protocol which have a balance sheet similar to a traditional lending institution. The lending market maintains a liquidity pool of supported assets. It raises capital by offering a interest bearing token to those who supply capital to the protocol. A user who wishes to borrow from the market posts a set of admissible tokens as collateral, and selects an asset to borrow. The ratio of the borrowed balance to the market value of the collateral is bounded by a protocol governance defined loan-to-value cap, and the threshold θ in the health-factor inequality above is the per-asset liquidation threshold applied to the collateral value.

In §6.4 we test whether the size of a lending market, as measured by total value locked, explains the market's response to a common shock to the price of the collateral asset. This is motivated by the fact that the TVL of a lending pool is a direct proxy for its market depth: a larger pool can absorb a given dollar volume of forced collateral sales over a wider book of suppliers and borrowers, and the marginal liquidation moves the position of the pool by less. Alternatively, in a shallow pool the same forced sale moves a larger fraction of the book, pushes more borrowers across their liquidation thresholds, and amplifies the cascade. Under this depth-insulation prior, deeper markets should produce smaller liquidation responses to a given collateral-return shock (Gudgeon et al., 2020).

3.3 Heterogeneity and the Cross-Sectional Unit

The institutional features above vary across protocol–chain pairs, and the dynamics of the liquidation mechanism vary with them. Specifically, at the protocol level, pools differ in whether collateral is shared across a single cross-collateral pool or isolated into per-pair markets, in the size of the liquidation bonus paid to liquidators, in the close factor that bounds how much of a position can be seized in a single liquidation, in the per-asset loan-to-value cap and liquidation threshold, and in whether liquidations are atomic fixed-spread sales or descending-price auctions. At the chain level, the same protocol code deployed on different blockchains faces different supplier and borrower populations, different gas regimes, different block times, and a different set of off-protocol venues against which seized collateral can be unwound. Each protocol–chain pair therefore carries its own borrow book, its own supply-side liquidity, and its own off-protocol exit liquidity for the assets it accepts as collateral, so liquidity in the sense relevant to cascade propagation is distinct at the protocol–chain level rather than at the protocol level alone. The cross-sectional unit (CSU) for the empirical analysis is therefore defined as a unique protocol–blockchain pair:



This definition captures both sources of heterogeneity in a single index and motivates the econometric framework, which estimates separate dynamics for each unit.

4 Data

4.1 Variable Definitions

4.1.1 Liquidation

Liquidation is defined as the USD value of collateral seized by liquidators (or absorbed by the protocol, depending on design) during a daily interval, transformed as $\ln(1 + x)$ to accommodate zero-liquidation days and reduce skewness. This measure follows Lehar & Parlour (2022) and OECD (2023), who similarly aggregate daily liquidation volume in USD.

4.1.2 Collateral Basket Return

For each collateral asset i in a CSU, define the daily log return $r_{i,t} = \log(p_{i,t}) - \log(p_{i,t-1})$. Let $w_{i,t-1}$ denote the share of asset i in the collateral basket at $t-1$. The collateral basket return is

$$R_t = \sum_i w_{i,t-1}\, r_{i,t},$$

which is the stationary I(0) series that enters the SVAR alongside $\log(1 + \text{liquidation})$.

4.1.3 Total value locked (TVL)

For each CSU i and day t, the supply-side total value locked is the sum across supported assets of the USD value of tokens deposited in the protocol,

$$\text{TVL}_{i,t}^{\text{supply}} \;=\; \sum_{j \in \mathcal{A}_i} P_{j,t}^{C} \cdot Q_{j,t}^{\text{supply}},$$

where $\mathcal{A}_i$ is the set of supported collateral assets, $Q_{j,t}^{\text{supply}}$ is the protocol-level supplied quantity of asset j read from on-chain state at daily close, and $P_{j,t}^{C}$ is the same oracle price used by the protocol for that asset. For cross-sectional analysis, each CSU's size is summarized by the sample-window mean of the supply-side series, $\overline{\text{TVL}}_i = T^{-1} \sum_t \text{TVL}_{i,t}^{\text{supply}}$; $\log(\overline{\text{TVL}}_i)$ is the regressor used in the second-stage tests of §6.4.

4.2 Data Pipeline

Assembling a balanced protocol–chain panel of liquidations, collateral deposits, and oracle prices from raw blockchain state is non-trivial: each protocol has its own contract architecture, event signatures, and accounting schema, and the data spans seventeen blockchains. Existing approaches face documented limitations—Tian & Zhu (2025) excluded all Compound liquidations from their Bank of Canada study due to decoding errors in the commercial source, subgraph indexing frequently breaks, and most academic papers build ad hoc scripts for a single protocol on a single chain. We assemble an open-source pipeline organized around a protocol-adapter pattern and a three-layer Bronze/Silver/Gold transformation, which produces the analysis-ready panels used below. A full architectural description—design principles, adapter interface, RPC economics, scale, and extensibility—is given in Appendix B.

4.3 Sample Description

Coverage includes: Aave V2/V3, Compound V2/V3, Morpho Blue, Euler V2, Fluid, SparkLend, Venus, BenQi, Moonwell, LayerBank, Lodestar, Mendi, ZeroLend, and Keom—on Ethereum, Arbitrum, Base, Optimism, Polygon, Avalanche, Binance, Gnosis, Scroll, Linea, zkSync, Celo, and Sonic.

Table 1: Sample Composition

	Bivariate (liquidation, basket return)
Variables	Liquidation, Basket Return
Pipeline protocol adapters	23
Pipeline blockchains	17
CSUs estimated	32
Date range	July 2024–December 2025

5 Econometric Methodology

5.1 Pedroni (2013) Heterogeneous Panel VAR

The econometric framework follows Pedroni (2013). For each cross-sectional unit $i \in \{1, \ldots, N\}$, a reduced-form VAR(p) is estimated:

$$\mathbf{y}_{i,t} = \sum_{\ell=1}^{p} \mathbf{R}_{i,\ell}\, \mathbf{y}_{i,t-\ell} + \mathbf{u}_{i,t} \tag{1}$$

where for the bivariate model:

$$\mathbf{y}_{i,t} = \begin{pmatrix} \mathrm{ret}_{i,t} \\ \mathrm{liq}_{i,t} \end{pmatrix}, \qquad \mathbf{u}_{i,t} = \mathbf{A}_i\, \boldsymbol{\mu}_{i,t}, \qquad \boldsymbol{\mu}_{i,t} = \begin{pmatrix} \mu_{i,t}^{\mathrm{ret}} \\ \mu_{i,t}^{\mathrm{liq}} \end{pmatrix}$$

where $\mathrm{ret}_{i,t}$ is the collateral basket log return and $\mathrm{liq}_{i,t}$ is the daily log-liquidation volume. $\mathbf{A}_i$ is the structural impact matrix. The matrices $\mathbf{R}_{i,\ell}$ are member-specific 2×2 coefficient matrices, and $\mathbf{u}_{i,t}$ is a 2×1 reduced-form residual vector.

Distinct from a pooled Panel VAR approach, every parameter is allowed to differ across units, accommodating the substantial heterogeneity across protocol–chain pairs documented in Section 3.1. Pesaran & Smith (1995) show that pooled estimators of the average coefficient under heterogeneous dynamics are inconsistent, so the heterogeneous panel structure is required for the kind of cross-sectional question the paper asks.

The estimation procedure follows the steps of Pedroni (2013, §3). *(i) Standardization.* Each variable is demeaned and rescaled to unit variance within CSU, so that members enter the cross-sectional aggregation on a common scale. Without this step, the panel mean is dominated by whichever CSU has the largest raw scale—log-USD liquidation across the panel ranges over several orders of magnitude (Section 4.3). *(ii) Member VARs.* A separate reduced-form VAR is fit to each CSU's standardized series with lag length chosen by AIC, and the identification described below is applied to recover member-specific structural shocks $\boldsymbol{\mu}_{i,t}$ from the reduced-form residuals. *(iii) Common reference VAR.* A single SVAR is fit to the panel-wide cross-sectional average $\bar{\mathbf{y}}_t = N^{-1} \sum_i \mathbf{y}_{i,t}$ under the same identification, yielding the common structural shock series $\bar{\boldsymbol{\mu}}_t$. Estimating the common shock from a separate aggregate-VAR rather than averaging member shocks ex post is what allows the common dynamics to differ from the typical member's—the panel-wide co-movement is a structural object in its own right, not a summary statistic over members. *(iv) Loadings.* Member-specific loadings $\lambda_{i,v}$ are estimated as the diagonal-element covariance between $\mu_{i,t}^{v}$ and $\bar{\mu}_t^{v}$, summarizing how much of member i's shock to variable v co-moves with the panel-wide common shock. Because both shock series are unit-variance after standardization, $\lambda_{i,v}$ is in $[-1, 1]$ and is interpreted as the correlation. The full member-specific impulse response decomposition then follows in Section 6.3.

We identify an interesting shock as one that moves the basket return contemporaneously and is thereby orthogonal to the same-day liquidation shock; equivalently, the liquidation shock is constrained to have zero same-day effect on the basket return. Schematically,

$$\mathbf{A}(0) = \begin{bmatrix} \cdot & 0 \\ \cdot & \cdot \end{bmatrix},$$

the lower-triangular Cholesky implements the orthogonalization. Without the orthogonalization the impulse responses reported below would trace dynamic effects of reduced-form residuals; with it, they trace the dynamic effects of two economically separate shocks.

The return shock. The return shock $\mu_{i,t}^{\text{ret}}$ is the innovation in the collateral basket return $\text{ret}_{i,t}$, and the orthogonalization places no restriction on its contemporaneous effect on liquidation. Economically, it represents the part of a day's collateral-basket movement that arrives from outside the lending market's own state—a price move the protocol has not itself caused. Examples include broad crypto-wide selloffs or token-specific news that re-prices a single collateral asset without same-day liquidation activity.

The liquidation shock. The liquidation shock $\mu_{i,t}^{\text{liq}}$ is the innovation in daily log-liquidation volume after lagged dynamics are absorbed, with the orthogonalization placing zero same-day effect on the basket return. Economically, it represents the part of a day's liquidation activity that does not trace back to a same-day price move on the collateral basket—liquidations driven by something other than today's market direction. Examples include operational backlog (liquidator bots clearing positions that became unhealthy on a prior day but were uneconomic to seize until gas costs fell or DEX depth recovered), protocol-specific solvency events (a wrapped-asset price feed marking down a single token while the rest of the basket is unchanged, an interest-rate model pushing utilization through a kink that erodes a borrower's health factor with no price move), and bad-debt cleanup or self-liquidation flows that run through the same liquidation pipeline without being driven by a contemporaneous return innovation.

5.2 Impulse Response Functions

Impulse response functions (IRFs) trace the dynamic effect of a structural shock to variable j on variable k over horizon h. Because the model is estimated separately for each unit, common and idiosyncratic IRFs are constructed as the median across members and measure the total accumulated effect.

Both structural shocks are normalized to a common reference-event scale before IRFs are reported, so that the panels can be read on comparable terms rather than on the unit-variance scale. The return shock is normalized so that its contemporaneous effect on the basket return at $h = 0$ equals -1% (a 1% same-day collateral-price decline). The liquidation shock is normalized so that its contemporaneous effect on liquidation at $h = 0$ equals a +1% same-day increase in USD liquidation volume. Normalization is applied per IRF type: the common and idiosyncratic decompositions are each scaled independently against their own $h = 0$ reference, so both hit the same target at $h = 0$ and can be compared horizon-by-horizon. The own-variable response at $h = 0$ is therefore mechanically equal to the normalization target for each shock and each IRF type; the economically informative quantities are the cross-shock responses and the $h \geq 1$ dynamics.

5.3 Second-stage cross-sectional regression

The cross-sectional object the second stage takes as its dependent variable is the common-component response at h=1, which measures the part of CSU i's +1 day liquidation response that co-moves with the panel-wide common shock. The second stage regresses this object on CSU-level characteristics and asks which features of a protocol–chain pair predict a strong or weak cascade response to a common collateral-price move.

Formally, for a CSU-level regressor x_i (e.g., $\log(\overline{\mathrm{TVL}}_i)$ for the depth test, the stablecoin share of the debt book for the denomination test), the second stage is an OLS of the form

$$\mathrm{IRF}^{\mathrm{common}}_{i,\,h=1} \;=\; a + b\,x_i + \epsilon_i, \qquad i = 1, \ldots, N,$$

estimated with HC3-robust standard errors across the CSUs for which clean moderator data are available—$N = 34$ for the depth test and $N = 33$ for the denomination test (which additionally requires at least 200 observed liquidation events for a reliable count-weighted stablecoin share). The design is deliberately simple: the first-stage SVAR has already done the within-CSU orthogonalization and absorbed the time-series dynamics; the second stage is purely cross-sectional.

6 Results

The Panel SVAR is estimated separately for each of the 38 qualified CSUs, and the member-specific impulse responses are decomposed into common and idiosyncratic components following Pedroni (2013). We describe the patterns across the panel; the underlying figures report the cross-member median alongside the interquartile range of responses.

6.1 Impulse Responses

Consider first the common-component responses in Figure 1. A return innovation normalized to a -1% basket-return decline at $h = 0$ produces an immediate rise of roughly +22% in USD liquidation volume on the same day. The response peaks at $h = 1$ at approximately +28% and decays back toward baseline within about five days. The interquartile band at $h = 0$ lies entirely above zero, indicating a same-day price-triggering channel

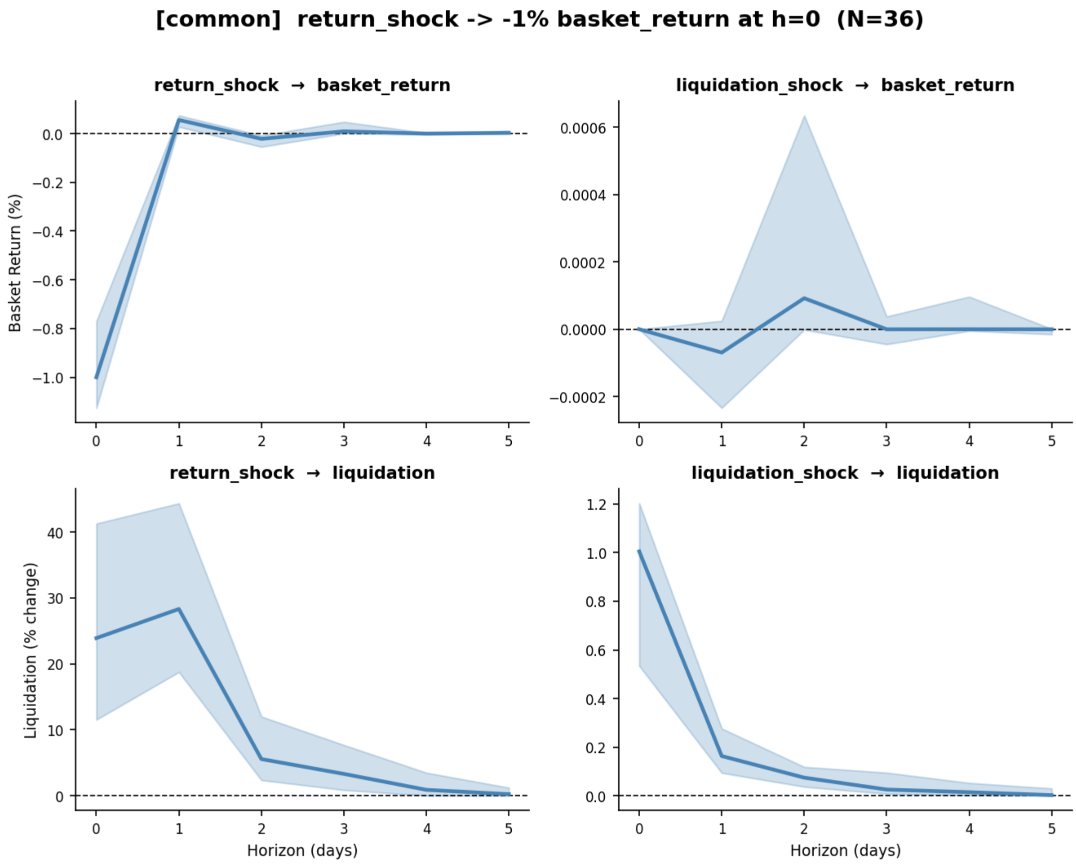


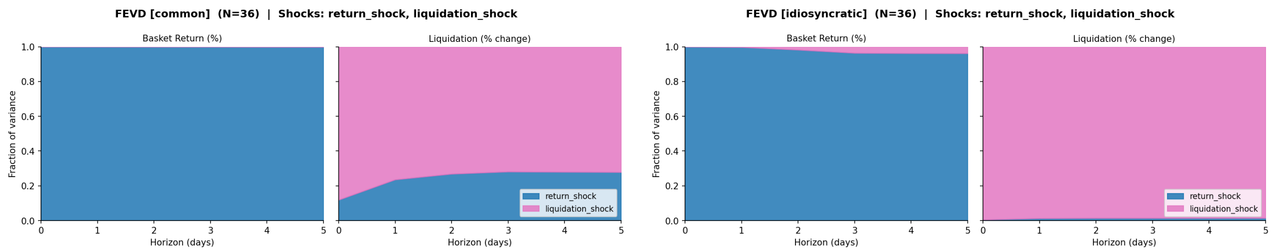
Figure 1: Common-component impulse responses. Each panel shows the cross-member median (solid) and interquartile range (shaded) of the response indicated. The return shock is normalized to a -1% basket-return response at $h = 0$.

that is consistently signed across the panel. The basket-return response to its own shock decays sharply from the normalized -1% at $h = 0$ and reverts within two days.

A liquidation innovation, by construction, has no contemporaneous effect on the basket return. One day later the return response is small and slightly negative—on the order of a few hundredths of one percent—which is the lagged forced-sale channel the VAR dynamics recover at $h \geq 1$. The liquidation shock's own-variable response is set to $+1\%$ at $h = 0$ by normalization, and decays rapidly thereafter—to roughly $+0.22\%$ at $h = 1$ and essentially zero by $h = 5$. The fast own-shock decay means that a given day's liquidation innovation does not persist meaningfully into the following days once controlled for return dynamics, which is consistent with the intra-block resolution of individual cascade episodes documented by Perez et al. (2021). The corresponding idiosyncratic-component responses follow the same qualitative shape and are reported in Appendix C; the cross-sectional heterogeneity that motivates the second-stage analysis is summarized below through the λ decomposition.

6.2 Variance Decomposition

Figure 2 summarizes the forecast-error variance decomposition at each horizon, separately for the common and idiosyncratic components. The return shock accounts for essentially all of the variance in the basket return at



(a) Common component

(b) Idiosyncratic component

Figure 2: Forecast-error variance decomposition at each horizon, common (left) and idiosyncratic (right) components.

every horizon—100% at $h = 0$ by construction of the orthogonalization, and 99% or more at longer horizons. The liquidation shock accounts for the bulk of its own-variable variance, roughly 97% at $h = 0$, falling to about 92% by $h = 5$ as the return shock picks up a small share (around 7%–8%).

6.3 Common versus Idiosyncratic Loadings

Table 2 reports the cross-member median of the diagonal λ loadings.

Table 2: Median λ across 38 CSUs.

Variable	Median λ
Basket return	0.90
Liquidation	0.50

The two variables load on the common component very differently. For the basket return, the median λ is high: most of the variation in collateral returns across CSUs co-moves with the panel-wide common factor. This is broadly consistent with the observation that collateral baskets across protocols and chains overlap heavily in their exposure to a small number of crypto assets. For liquidation, the median λ is substantially lower. A large portion of the variation in daily liquidation volume is member-specific.

6.4 Cross-sectional heterogeneity in the liquidation response

The member-specific impulse responses and the λ loadings describe a panel in which daily collateral-return dynamics are heavily common across CSUs while daily liquidation dynamics are substantially less so. A natural next question concerns what predicts this cross-sectional heterogeneity. Following the spirit of Pedroni (2013), we treat the per-member responses as objects of interest in their own right and use them as dependent variables in a second stage. Two candidate moderators suggested by the institutional structure of DeFi lending take up the remainder of this section: the depth of the lending market, measured by total value locked, and the denomination of the debt being borrowed, measured directly through the stablecoin share of each CSU's debt book.

6.4.1 Market depth

A common intuition in discussions of DeFi lending is that deeper markets should be insulated from cascade dynamics. More supply-side liquidity means more room to absorb forced sales, more heterogeneous borrowers, and more distance between aggregate positions and liquidation thresholds. Under this view, the per-CSU price-triggering response should decline in market size: a -1% collateral-return shock should produce a smaller absolute liquidation response in a deep market than in a shallow one.

We test this hypothesis through the second-stage construction recommended in Pedroni (2013): regress the per-CSU common-component impulse response on a CSU-level observable. The dependent variable is the common-component response at h=1 of liquidation to a return shock, $\mathrm{IRF}^{\mathrm{common}}_{i,h=1}$. The regressor is $\log(\overline{\mathrm{TVL}}_i)$, the log of mean daily supply-side total value locked over the sample window. We estimate

$$\mathrm{IRF}^{\mathrm{common}}_{i,h=1} \;=\; \alpha + \beta \cdot \log(\overline{\mathrm{TVL}}_i) + \varepsilon_i,$$

with HC3-robust standard errors across the 34 CSUs for which clean TVL data are available.

Table 3: Depth test: bivariate cross-sectional regression of $\mathrm{IRF}^{\mathrm{common}}_{i,h=1}$ on $\log(\overline{\mathrm{TVL}}_i)$. HC3-robust standard errors.

	$\hat{\beta}$	SE	t	p	R^2	N
$\log(\overline{\mathrm{TVL}}_i)$	-0.008	0.007	-1.10	0.27	0.03	34

The result is null, as reported in Table 3. The slope is, if anything, weakly negative–meaning slightly more cascade response in larger markets, contrary to the depth-insulation prior–but it is not distinguishable from zero. We cannot reject the hypothesis that depth has no effect on the cross-sectional magnitude of the price-triggering response.

Using the idiosyncratic decomposition rather than the common component yields slopes that are also indistinguishable from zero at $h = 1$. Depth, in this panel, does not predict the magnitude of the price-triggering response at h=1 in any of the standard ways the question can be asked.

Two readings are consistent with this null. The first is that the depth-insulation prior is empirically irrelevant: in DeFi lending, a -1% collateral move triggers a roughly comparable cascade response across markets regardless of pool size, because the cascade mechanism (oracle update breaching health-factor thresholds) is largely independent of how much liquidity sits behind the borrow book. The second is that depth does have a real effect that we are not powered to detect: with $N = 34$ and cross-sectional noise in the first-stage IRF estimates, modest size effects could be obscured by sampling variation. We do not take a position between these readings; what the panel can say is that the simplest test of depth-insulation does not survive at conventional significance levels.

6.4.2 Stablecoin share of the debt book

A second candidate moderator is the denomination of the debt being borrowed. In a lending position with collateral C, debt D, and liquidation threshold θ, the health factor $\mathrm{HF} = \theta \cdot P^C Q^C / (P^D Q^D)$ reduces in log changes to the difference between the collateral return and the debt return, absent quantity and rate effects. Under stablecoin debt the debt return is effectively zero, and broad crypto-price movements pass through to the health factor unimpeded. Under debt denominated in an asset correlated with collateral, the same broad move leaves the health factor comparatively stable. A stablecoin-denominated borrow book therefore concentrates cascade-triggering exposure.

For each CSU we classify every liquidation event by the debt token repaid, using a canonical stablecoin list available at github/...... (USDC, USDT, DAI, GHO, FRAX, LUSD, and their bridged and wrapped cToken / vToken variants). The stablecoin share is the count-weighted fraction of a CSU's liquidation events whose debt token is classified as a stablecoin,

$$\mathrm{StableShare}_i \;=\; \frac{1}{n_i} \sum_{\ell \in \mathcal{E}_i} \mathbf{1}\{\mathrm{Stable}(j_\ell^D)\},$$

bounded in $[0, 1]$.

To keep the share well-identified we restrict the second stage to CSUs with at least 200 liquidation events; further requiring clean TVL data and a converged SVAR estimate yields a working sample of 33 CSUs.

The second-stage regression is

$$\mathrm{IRF}_{i,h=1} \;=\; \alpha + \beta \cdot \mathrm{StableShare}_i + \varepsilon_i,$$

estimated with HC3-robust standard errors, where $\mathrm{IRF}_{i,h=1}$ is the common-component response of liquidation to a normalized -1% return shock. Under the HF microfoundation, the signature prediction is $\hat{\beta} < 0$: markets with a higher stablecoin-debt share should respond more sharply (a more negative IRF, since the response direction is negative) to a given collateral-price decline.

Table 4 reports the main estimates. The slope on stablecoin share is $\hat{\beta} = -0.306$ (HC3 SE 0.044, $t = -6.99$, $R^2 = 0.54$): moving from a 0% to a 100% stablecoin-debt book corresponds to about 36% additional USD-liquidation per -1% return shock.

Table 4: Second-stage regressions of $\mathrm{IRF}_{i,h=1}$ on stablecoin-debt share. HC3-robust standard errors; $N = 33$ CSUs with ≥ 200 liquidation events, clean TVL data, and a converged first-stage SVAR.

	$\hat{\beta}_{\mathrm{StableShare}}$	SE	t	R^2
$\mathrm{IRF}_{i,1}^{\mathrm{common}}$	-0.306	0.044	-6.99	0.54

The bivariate plot in Figure 3 shows the cross-section directly. The stablecoin cluster at the right of the panel—Compound V3 markets on Ethereum, Arbitrum, and Base, together with SparkLend on Ethereum—sits

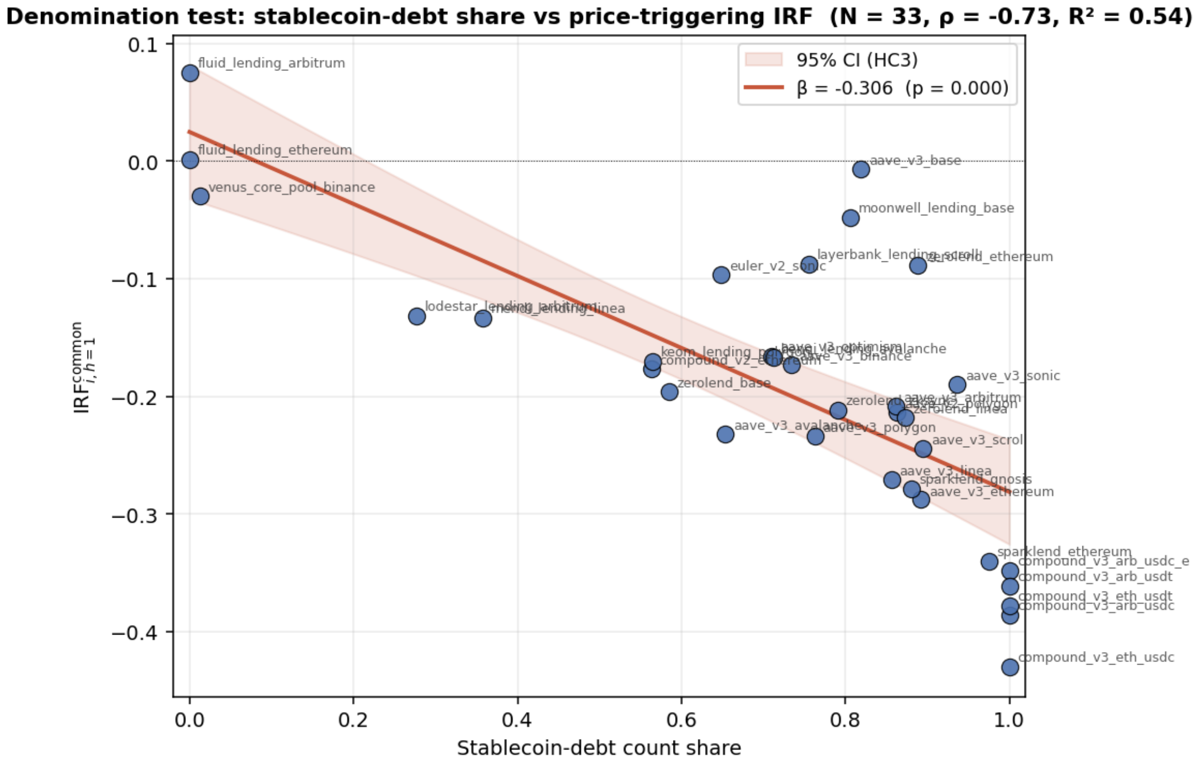


Figure 3: Bivariate cross-sectional regression of the common-component IRF at $h = 1$ on the count-weighted stablecoin-debt share. Each point is one CSU; the fit line is HC3-robust OLS.

visibly below the zero line, while the lowest-share markets (Fluid deployments, Venus core pool on Binance) sit near zero or marginally above it. The ordering from left to right traces the HF microfoundation: as the stablecoin share rises, the offsetting debt-return term in $\Delta \log(\mathrm{HF}) = r^C - r^D$ drops out, more of the full collateral return flows into the health factor, and more positions cross the liquidation threshold for a given 1% move.

7 Discussion

Before turning to the cross-sectional tests, note the magnitude of the panel-wide common response itself: a -1% same-day shock to the collateral basket return triggers a roughly $+22\%$ rise in USD liquidation activity on the same day, peaking at about $+28\%$ at $h = 1$ and decaying back to baseline within five days. The price-triggering channel is therefore an economically substantial feature of the average DeFi lending market, not just a statistically detectable one, and the cross-sectional regressions below are about how that average response is distributed across protocol–chain pairs.

The two cross-sectional tests in §6.4 paint an asymmetric picture. The concept of a "deeper market", as represented by TVL, has no liquidation dampening effect in this panel. The denomination test, by contrast, directly recovers the HF microfoundation prediction: per-CSU price-triggering IRFs at $h = 1$ are tightly negatively related to the stablecoin share of the debt book, with $\hat{\beta} = -0.31$ and $R^2 = 0.54$. Cross-sectional cascade exposure in DeFi lending therefore appears to concentrate along the denomination dimension—stablecoin-debt books load more sharply on the price-triggering channel than volatile-debt books—without any clear corresponding gradient in market depth.

Read against the prior empirical literature, this is a sharper cross-sectional statement than has been available. Lehar & Parlour (2022) establish that liquidation-induced price impacts are substantial on average across protocols; Heimbach & Huang (2024) show that daily liquidation activity is event-driven rather than leverage-driven; Chiu & Danisman (2026) decompose the health-factor driver of the liquidation decision and isolate the collateral-return term. What these papers do not address is cross-sectional heterogeneity: given that some markets experience much more cascade pressure than others, which features of a market explain that heterogeneity? In the emerging-market context, dollar-denominated sovereign debt concentrated currency-crisis exposure because a depreciation of the local currency against the dollar moved the liability but not the local-currency revenue stream, leaving the sovereign's debt-service-to-revenue ratio exposed to exchange-rate moves rather than local fundamentals. In DeFi lending, stablecoin-denominated debt concentrates crypto-wave exposure because a broad crypto decline moves the collateral value but not the dollar liability, leaving the health factor exposed to the common crypto factor rather than to the idiosyncratic protocol state. The mechanism is mechanically the same, and the cross-sectional prediction—that exposure concentrates where liability denomination severs the offsetting return—is the same.

These findings should be read with appropriate caution. The sample covers a single 18-month window (July 2024 – December 2025) that spans an active rally and several drawdowns but no full bear-market stress regime, and replication on a longer horizon would help establish whether the denomination slope is stable across market regimes. TVL captures supplied liquidity inside the lending protocol but is a noisy proxy for the depth that cascades actually run through—the off-protocol DEX exit depth for the seized collateral—so the depth null may partly reflect mismeasurement of the relevant depth concept rather than the absence of an underlying size effect. The daily frequency itself aggregates intra-block dynamics that Perez et al. (2021) have shown to matter at shorter horizons. Finally, the two cross-sectional tests are small-N regressions ($N = 34$ for the depth test, $N = 33$ for the denomination test): the denomination slope is robust to clustering and control choices, but the size null is consistent both with the depth-insulation prior being economically irrelevant and with our being underpowered to detect a modest size effect.

8 Conclusion

This thesis argues, and empirically supports, one claim and reports a related null. At the panel-wide level, the common-component IRF puts a magnitude on the price-triggering channel: a -1% same-day collateral-return shock raises USD liquidation volume by roughly $+22\%$ on the same day and about $+28\%$ at $h = 1$ before decaying back to baseline within five days. The claim is that cascade exposure in DeFi lending is not randomly distributed across protocol–chain pairs: it concentrates along the denomination dimension, with stablecoin-debt markets responding more sharply to a -1% collateral-return shock than volatile-debt markets.

The mechanism follows directly from the health-factor microfoundation: stablecoin debt severs the offsetting debt-return term that would otherwise absorb part of a collateral-price decline. A cross-sectional regression of the per-CSU common-component IRF at $h = 1$ on the stablecoin share of the debt book recovers the predicted slope: roughly 36% additional USD-liquidation per -1% return shock when moving from a fully volatile-debt to a fully stablecoin-debt book. The corresponding test on market depth is null: regressing the same per-CSU IRF on log(TVL) yields a slope statistically indistinguishable from zero, so the depth-insulation prior fails to find empirical support but neither does the data show an opposing concentration of exposure with size.

A portion of the contribution is methodological. The analysis is supported by an open-source multi-chain data pipeline covering 23 protocol families across 17 blockchains, which we hope will be useful to subsequent work on DeFi lending. Natural extensions include widening the set of cross-sectional moderators (protocol architecture, oracle type, liquidation-mechanism design), moving to block-level or hourly data where the cascade dynamics can be resolved more finely (Perez et al., 2021), and testing whether the stablecoin debt pattern is stable out of sample when the stablecoin universe itself expands, diversifies, or is disrupted.

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Additional Background Reading

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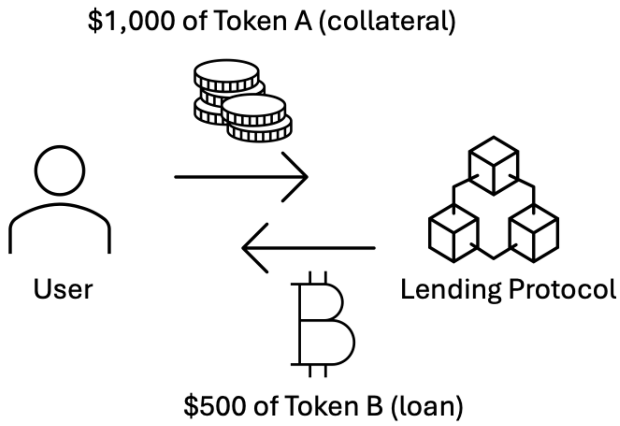
A Leverage, Liquidation, and Volatility Amplification: A Numerical Example

A.1 Leverage

A user deposits \$1,000 of Token A as collateral and borrows \$500 of Token B, given a maximum loan-to-value (LTV) of 50%. After one cycle, the user has exposure to \$1,500 worth of cryptocurrency. To amplify returns on Token A, the user can convert the borrowed Token B back into Token A and deposit it as additional collateral:



If this process is repeated, collateral follows a geometric series converging to $\frac{\text{Equity}}{1-\text{LTV}} = \frac{1{,}000}{1-0.5} = \$2{,}000$, yielding leverage of $2\times$ in the limit. With a liquidation threshold $\theta = 70\%$ and the health factor defined as

$$HF = \frac{\text{Collateral Value} \times \theta}{\text{Debt}},$$

the initial health factor is $HF_0 = \frac{2{,}000 \times 0.70}{1{,}000} = 1.4$. Liquidation occurs when $HF \leq 1$, requiring a price decline of

$$\Delta = 1 - \frac{1}{HF_0} = 1 - \frac{1}{1.4} \approx 0.29.$$

At $2\times$ leverage, a price decline of only 29% triggers liquidation.

A.2 Liquidation

Consider a 30% price decline. Collateral falls from \$2,000 to $2{,}000 \times 0.70 = \$1{,}400$. A liquidator repays the full \$1,000 debt in exchange for collateral worth $1{,}000 \times 1.05 = \$1{,}050$ (including a 5% liquidation bonus β). The borrower retains $1{,}400 - 1{,}050 = \$350$—a loss of \$650 on \$1,000 of equity.

A.3 Volatility Amplification

Without leverage, a 30% decline reduces the position by 30%. Under 2× leverage, the same decline produces a loss of $1{,}000 - 350 = \$650$, or 65% of equity. At the system level, a 2× levered position requires twice as much collateral to be liquidated as an unlevered one for the same price shock. Because liquidation flows enter the market as sell pressure, higher leverage mechanically increases the quantity of collateral sold per unit of price movement.

B Data Pipeline Architecture

The pipeline is a set of configurable infrastructure components—protocol adapters, chain configurations, collection scripts, and transformation layers—designed to be extended without modifying existing code. The full codebase is available on GitHub; users supply their own RPC provider API keys.

B.1 The Data Challenge

Each lending protocol implements a different smart contract architecture with its own event signatures and encoding conventions. A liquidation on Aave V3 emits a single `LiquidationCall` event; the same action on Compound V3 emits two separate events (`AbsorbDebt` and `AbsorbCollateral`) that must be joined by transaction hash. Token amounts are raw integers requiring decimal normalization, and USD conversion requires historical on-chain price data. DeFi spans 15+ blockchains, each requiring separate RPC access and chain-specific middleware. Prior approaches—Dune Analytics, The Graph, commercial providers, and ad hoc scripts—each face constraints summarized in Table 5.

Table 5: DeFi Data Approaches in Prior Research

Approach	Used by	Limitation
Dune Analytics	Tian & Zhu (2025)	Excluded Compound data (decoding errors)
Kaiko (commercial)	OECD (2023)	\$10K–55K/year; enterprise contracts
Ad hoc scripts	Qin et al. (2021); Lehar & Parlour (2022)	Single protocol, single chain; not released
The Graph	Various	Subgraphs break; one per protocol per chain

B.2 Pipeline Design

The pipeline is organized around *protocol abstraction* and *layered transformation*.

Protocol abstraction. Each protocol implements a self-contained adapter with three methods: (1) `resolve_market()`—discover contract addresses from a registry; (2) `fetch_events()`—query raw event logs via `eth_getLogs`; (3) `normalize()`—map protocol-specific fields to a unified schema. All protocol-specific knowledge (event signatures, ABI layouts, multi-event joins) is encapsulated within the adapter. Adding a new protocol requires ≈100 lines of Python; adding a new chain requires one YAML entry. Adapters exist for 23 protocol families

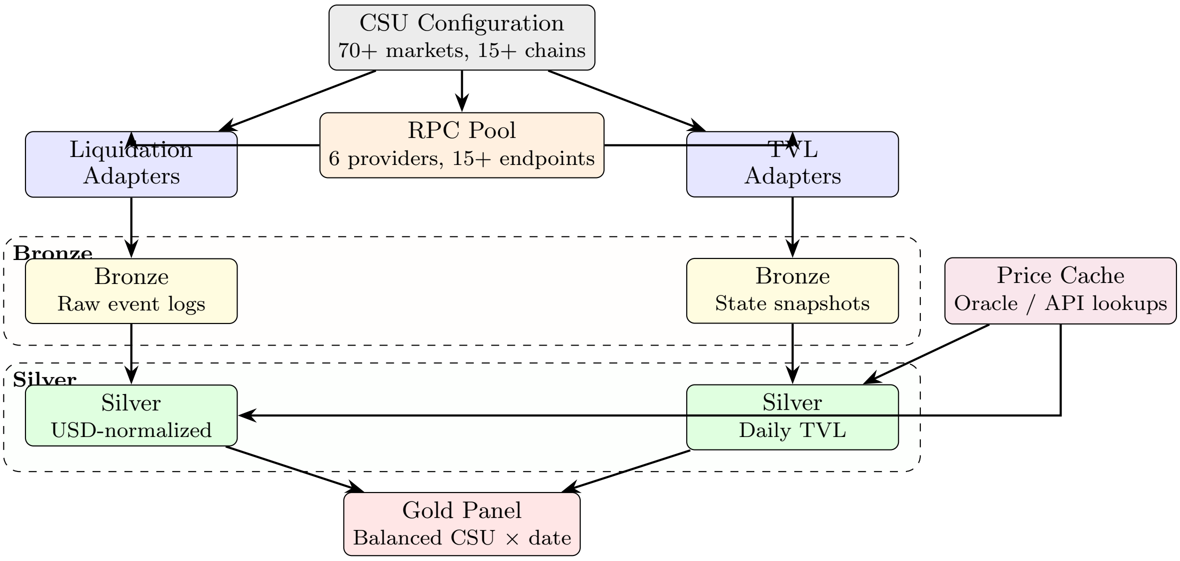


Figure 4: Data pipeline architecture. A YAML configuration drives collection across all protocols and chains. Protocol-specific logic is encapsulated in adapters; downstream layers operate on unified schemas. The price cache feeds USD conversion at the silver layer. Each layer is independently verifiable and resumable from checkpoints.

(Aave V2/V3, Compound V2/V3, Morpho Blue, Euler V2, Fluid, SparkLend, Venus, BenQi, Moonwell, LayerBank, Lodestar, Mendi, ZeroLend, Keom, and others). The same adapter runs identically across all chains where a protocol is deployed.

Layered transformation. Raw blockchain data passes through three layers (Figure 4):

1. **Bronze.** Raw event logs and state snapshots stored as append-only JSONL, bit-for-bit reproducible from the chain. Checkpoint files enable incremental, resumable collection.

2. **Silver.** Unified-schema Parquet files with token metadata enrichment, block timestamp resolution, USD conversion via a price cache built from on-chain oracle queries, and protocol-specific edge-case resolution (e.g., Compound V3's two-event join).

3. **Gold.** Balanced CSU $\times$ date panels with derived variables: utilization, $\log(1 + \text{collateral seized})$, and rolling volatility.

Unified collection. A single `eth_getLogs` call per chain block range uses OR-ed topic signatures to scan for liquidation events across all protocols simultaneously, dispatching each event to the appropriate adapter.

RPC routing. A request-routing layer distributes calls across six provider tiers with per-provider rate limiting, exponential backoff, usage tracking, and automatic key blacklisting. One panel row requires $\approx$860 RPC calls per CSU per day; the full scope ($\sim$42 CSUs, 14 chains, 2 years) requires an estimated 15 million smart contract calls.

B.3 Extensibility

The pipeline can be extended along three dimensions without modifying existing code:

1. **New protocols.** Write one adapter (≈100 lines) implementing the three-method interface.

2. **New chains.** Add a YAML entry with protocol, chain, and contract address.

3. **New variables.** Modify the gold-layer script to derive different panel variables from the same silver inputs.

C Idiosyncratic-Component Impulse Responses

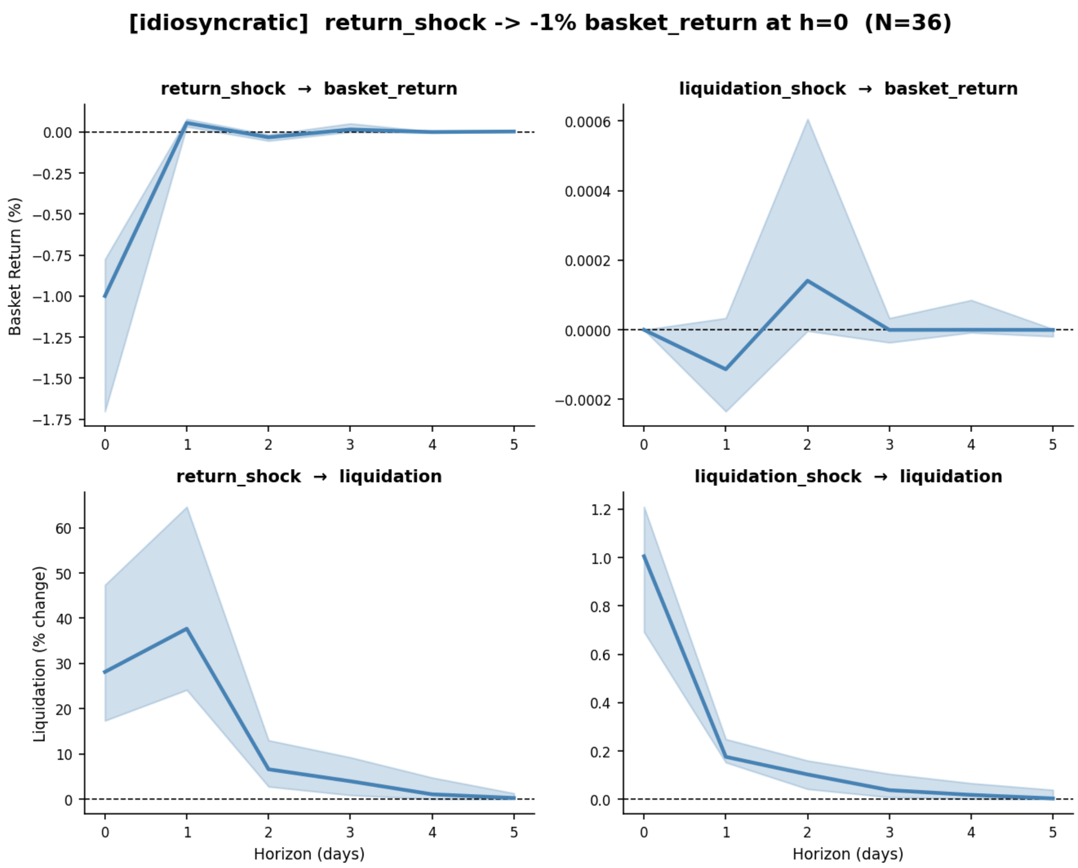


Figure 5: Idiosyncratic-component impulse responses. Each panel shows the cross-member median (solid) and interquartile range (shaded) of the response indicated.